GSDFuse: Capturing Cognitive Inconsistencies from Multi-Dimensional Weak Signals in Social Media Steganalysis

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arXiv:2505.17085v1 [cs.CR] 20 May 2025

Abstract

The ubiquity of social media platforms facilitates malicious linguistic steganography, posing significant security risks. Steganalysis is profoundly hindered by the challenge of identifying subtle cognitive inconsistencies arising from textual fragmentation and complex dialogue structures, and the difficulty in achieving robust aggregation of multi-dimensional weak signals, especially given extreme steganographic sparsity and sophisticated steganography. These core detection difficulties are compounded by significant data imbalance. This paper introduces GSDFuse, a novel method designed to systematically overcome these obstacles. GSDFuse employs a holistic approach, synergistically integrating hierarchical multi-modal feature engineering to capture diverse signals, strategic data augmentation to address sparsity, adaptive evidence fusion to intelligently aggregate weak signals, and discriminative embedding learning to enhance sensitivity to subtle inconsistencies. Experiments on social media datasets demonstrate GSDFuse's state-of-the-art (SOTA) performance in identifying sophisticated steganography within complex dialogue environments. The source code for GSDFuse is available at https://anonymous.4open.science/r/GSDFuse-B1E7.

1 Introduction

Steganography, the art of covert communication by embedding secret data within innocuous carriers (Cox et al., 2007), presents a dual-use dilemma, vital for privacy, yet a potent tool for illicit activities like cyberattacks and disinformation (Bieniasz and Szczypiorski, 2018). As steganographic techniques grow in sophistication, robust steganalysis becomes crucial for safeguarding digital ecosystems.

Text's ubiquity, malleability, and persistence in information sharing establish it as a prevalent steganographic medium. The rise of social media, with its vast user base and high volume of interactive content, has further amplified text's utility as a covert communication channel (Li et al., 2018), creating ideal environments for steganography. Modern linguistic steganography is increasingly dominated by generative approaches (Yang et al., 2018, 2020a). These methods leverage language models to generate cover texts, embedding information by subtly altering token probabilities during generation. Notably, recent advancements have pushed towards provably secure steganography (Zhang et al., 2021; Ding et al., 2023), where the conditional probability distribution of steganographic texts becomes virtually indistinguishable from that of benign texts. Offering high embedding capacity and flexibility in cover generation, these methods also achieve great imperceptibility.

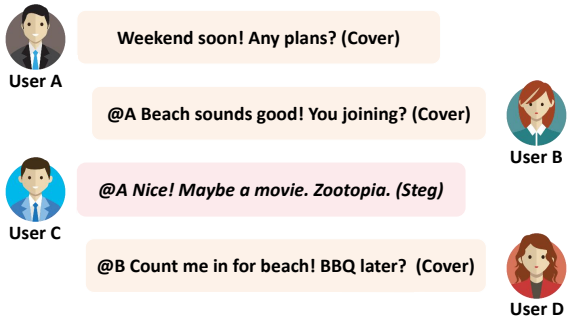

Figure 1: An illustrative example of a social media dialogue tree. A steganographic message (Steg, User C) is subtly embedded among benign messages (Cover). This scenario highlights key challenges such as textual fragmentation, steganographic sparsity, and complex message interactions that contribute to *systemic distortions* and *latent multi-faceted imperceptibility* in social media steganalysis.

Steganalysis, crucial for maintaining the security of public cyberspace, aims to find differences between steganographic and cover texts. Early deep learning models showed promise analyzing isolated

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texts (Yang et al., 2019b; Wu et al., 2021; Zou et al., 2020). However, social media's fragmented dialogues (e.g., short posts, threaded replies) severely limit extractable statistical features from single messages. This greatly hinders reliable detection based only on isolated text statistics, especially as steganography nears perfect statistical imperceptibility. This challenge drove a key shift in steganalysis: moving from breaking single-text statistical imperceptibility to using broader context to spot cognitive anomalies (Yang et al., 2021a,b). The core idea is that isolated benign-looking messages, when viewed within larger conversational or relational structures, can reveal subtle, "unnatural" patterns. Initial progress involved integrating immediate context (Yang et al., 2022). Later work further leveraged network topology and complex feature interactions, such as by modeling connection awareness (Pang et al., 2023) or using advanced graph architectures with attention for richer context from dialogue structures (Lu et al., 2025).

Despite these advancements in leveraging context, two profound challenges remain central to robust steganalysis in social media dialogues. Firstly, modern steganography increasingly aims for **Cognitive Imperceptibility**, especially within the fragmented and interactive nature of online conversations, illustrated in Figure 1. This means steganographic signals are designed to be indistinguishable from benign content, not just statistically at a local level, but also in terms of their naturalness and coherence within the broader dialogue flow and relational patterns. Detecting such deeply embedded content requires moving beyond surface-level features to identify subtle cues that disrupt this cognitive consistency, often only apparent when assessing the overall contextual fabric. Secondly, the difficulty of breaking this cognitive imperceptibility is compounded by the need to **Aggregate Multi-dimensional Weak Signals**. Steganography disperses faint steganographic traces across various dimensions, from deep semantic nuances and local message interactions to global dialogue structures. Effectively identifying steganography thus requires a method capable of not only constructing contextual abstracts but also of synthesizing these often sparse and individually inconclusive signals from multiple sources into a coherent judgment. Addressing these intertwined challenges is the primary motivation for our work.

The challenge of detecting steganography in social media dialogues is two-fold: while steganography achieves remarkable perceptual and statistical imperceptibility, subtle yet detectable **Cognitive Inconsistencies** can emerge within the complex conversational context. However, exploiting these inconsistencies requires overcoming the difficulty of **Aggregating Multi-dimensional Weak Signals** that indicate their presence. To address this, this paper introduces the following contributions:

- We propose **GSDFuse**, a novel, multi-component collaborative method designed to exploit subtle cognitive inconsistencies and enable robust aggregation of multi-dimensional weak signals by systematically extracting and adaptively fusing diverse features to identify deeply embedded, multi-dimensional anomalies within social media dialogue trees.
- The proposed GSDFuse method systematically dismantles this multi-faceted challenge by synergistically integrating hierarchical feature representation from semantic node profiling to topological contextualization, adaptive cross-modal feature fusion, discriminative embedding optimization, and robust learning strategies for imbalanced and sparse data.
- Experiments on large-scale, real-world social media datasets (Reddit, X [Twitter], Weibo) against mainstream steganographic algorithms (AC, HC, ADG) at various embedding rates and sparsity levels demonstrate that GSDFuse achieves SOTA performance.

2 Related Works

Linguistic steganalysis aims to detect steganography by identifying distinctions between steganographic and cover texts. Initial methods (Chen et al., 2011; Xiang et al., 2014) relied on aggregating statistical cues from n-grams or lexical features. The rise of generative steganography, producing more natural texts, spurred a shift towards deep neural networks for feature extraction from individual texts (Yang et al., 2019a; Zou et al., 2020; Yang et al., 2019b, 2020b; Peng et al., 2023). However, these models just struggled with statistical imperceptible, but steganography evolved to achieve statistical and perceptual naturalness. To address this, Yang et al. (2023) proposed leveraging external knowledge for recognizing cognitively subtle steganography, highlighting the limitations of purely internal textual features. Recognizing

these limitations, a crucial paradigm shift towards context-aware steganalysis has emerged. Early efforts by Yang et al. (2022) initiated this shift by highlighting the necessity of integrating contextual information with textual features to move beyond isolated text analysis. Subsequently, to better harness relational data, Pang et al. (2023) focused on enhancing "connection-awareness" through graph representation learning and introduced interaction-based mechanisms for more effective fusion of textual and graph-derived features, addressing the challenge of simply concatenating disparate information. Recently, to tackle the intrinsic limitations of standard GNNs in processing complex graph-text data, Lu et al. (2025) incorporated Transformer and cross-attention mechanisms, aiming to refine feature aggregation from broader and more intricate social network contexts. These works represent significant steps towards more sophisticated contextual modeling in steganalysis.

Existing methods, while improving context capture, often fall short in holistically synthesizing faint, dispersed cues across diverse textual and structural dimensions, especially under extreme steganographic sparsity. This necessitates a more discriminative method capable of unveiling such elusive, context-dependent patterns, motivating GSDFuse.

3 Preliminaries and Methodology

3.1 Key Challenges in Linguistic Steganalysis

Modern linguistic steganography's primary goal is achieving both statistical and perceptual imperceptibility. Statistical Imperceptibility, defined by Cachin (2004) via KL divergence , $D_{KL}(P_C||P_S) = 0$ for perfect security. Furthermore, perceptual naturalness is as critical as statistical similarity. As Yang et al. (2020a) highlighted with the "Psic Effect", statistical closeness does not guarantee perceptual naturalness, and vice versa. This inherent tension means steganographic indicators frequently manifest as dispersed, weak signals across multiple linguistic feature facets. Consequently, effective steganalysis must not only identify these cognitive inconsistencies but also achieve robust aggregation of multi-dimensional weak signals.

3.2 Problem Formulation and Task Definition

We address steganalysis in social media dialogues, modeled as tree structures $\mathcal{T} = (\mathcal{V}, \mathcal{E})$. Here, $\mathcal{V} = \{v_1, \ldots, v_N\}$ is the set of N message nodes, each containing textual content x_i, and $\mathcal{E}$ represents the reply-to relationships between them. Each node v_i is associated with a binary label $y_i \in \{0, 1\}$, where $y_i = 1$ indicates a steganographic text (S) and $y_i = 0$ a benign cover text (C). Steganographic content is typically sparse within these dialogue trees.

The task is to develop a steganalysis model f that, for a given dialogue tree $\mathcal{T}$ and all its node texts $\{x_k\}_{k=1}^{N}$, predicts the label y_i for each node $v_i \in \mathcal{V}$. This is formulated as a node-level classification problem on the graph. For each v_i, the model f outputs probabilities $P(Y = S|v_i, \mathcal{T})$ and $P(Y = C|v_i, \mathcal{T})$. A node v_i is then classified as steganographic (S) if $P(Y = S|v_i, \mathcal{T}) > P(Y = C|v_i, \mathcal{T})$.

3.3 Overall Model Architecture

Our proposed GSDFuse, depicted in Figure 2, is a modular, multi-stage pipeline designed to systematically identify cognitive inconsistencies by effectively aggregating multi-dimensional weak signals inherent in social media steganalysis.

The **Hierarchical Feature Representation** module serves as the foundation, constructing initial node embeddings. Its primary goal is to capture both the nuanced semantic meaning of individual messages and their local relational context within the dialogue tree. This dual focus is crucial for initially exposing signals related to both perceptual naturalness and subtle structural deviations.

To address the broader, non-local nature of cognitive inconsistencies, the **Global Structural Contextualization** module complements these localized features. It derives graph-level or subgraph-level structural summaries, enabling the method to understand overarching topological patterns and anomalies that isolated node analysis would miss.

The **Adaptive Feature Fusion** module is then tasked with intelligently integrating the diverse set of features, spanning semantic content, local interactions, and global structure. Recognizing that steganographic indicators often manifest as multi-dimensional weak signals that are subtle, dispersed, and require effective aggregation, this module adaptively weighs and combines information to highlight the most salient indicators of steganography.

Finally, two critical challenges are addressed: steganographic sparsity and the need for clear class separation. The **Data Augmentation for Imbalanced Learning** module employs techniques